

Observational Exercise in SH2402

Instructions

The goal of this exercise is to acquaint yourself with the constellations and the movement of the sky during the night as well as practice your skills in calculating positions of objects in the sky. You should also obtain some ability in choosing suitable equipment for observing various objects.

After completing this exercise you should be able to compute approximate positions of objects in the sky at different times and different locations, and to use this to determine which objects can be observed at a given time and what is the best time to observe a given object.

In your report you should describe how you have performed your observations, which methods you used, and what could be improved; as well as give detailed calculations of the positions of some objects as described below.

Part 1: Observations of the sky without a telescope

Observe the sky on at least two different nights. During one of the nights you should observe on at least two different occasions some hours apart.

Before the observation you might want to look at a star chart to find out which objects you can expect to observe. Star charts can be found on the internet. A couple of examples are www.skymaps.com, <http://www.skymaponline.net/>, but many more can be found if you search for star charts (if you find a good version, make a note on the course pages and let the others know!). You may also find planetarium software, such as Stellarium (www.stellarium.org) useful.

One piece of advice is to start your observing project as soon as possible. The weather in Stockholm is highly unreliable, so you should take the chance to observe whenever you can. Starting early also gives you a better chance of observing changes over time.

Another piece of advice is to let your eyes adapt to the darkness for around 20 minutes. You will then see much more. Avoid using a flash light or using a phone since it destroys your night sight (you could use a reddened flash light).

1. Make a logbook of your observations
 - a. Always make a note of the date, time, observing site, weather, equipment you use and other things you find worth noting.
 - b. When observing an object, record this in the logbook. Which object is it? Where in the sky is it (direction, height above the horizon)? Is it bright? Diffuse? Is there anything else interesting nearby? Which constellation does it belong to?

2. Start by identifying some familiar constellations and objects. Can you see any of the following? If so, where in the sky?
 - a. The Big Dipper
 - b. Orion
 - c. Sirius
 - d. Cassiopeia
 - e. The Pleiades
 - f. The Milky Way
3. Do you see any planets? Which ones? Where?
4. Do you see the Moon? Where? What is its phase? Make a sketch.
5. Has anything changed since your earlier observation(s)? What?

Part 2: Calculating positions of objects in the sky

Part 2.1: Review questions

To help you understand the calculations you will do later, you should first answer the questions below.

- 1) If a star has an hour angle of 6 h, in which direction can you find it?
- 2) When a star is on the observer's meridian, what is the local sidereal time?
- 3) What is the hour angle, t , of the vernal equinox, Υ ?
- 4) What is the time called since an object passed the local meridian?
- 5) When is the sidereal time $\Theta = 0^{\text{h}} 0^{\text{m}} 0^{\text{s}}$?
- 6) Where in the sky can you find the Sun at noon?
- 7) What is the declination of Polaris?
- 8) Stockholm is located 3° east of the Swedish time meridian. This means that a star is in the South
 - a. At the same time as on the time meridian
 - b. Earlier
 - c. Later
- 9) If a star has declination 30° , what is its height above the horizon at its highest point? In which direction can you find it then?
- 10) In which direction of the sky can you find a star at its lowest point?

Part 2.2: Find out what can be observed

The goal of this exercise is to find out before an observation which objects can be expected to be visible at the time of the observation. In the following you should assume that the observation takes place on February 14 at 18.00, and that the observing site has the position $59^{\circ}21' \text{ N}$, $18^{\circ}0' \text{ E}$ (AlbaNova, Stockholm).

We will focus on determining the visibility of a few interesting objects, which are listed in the table on the next page. For each of these objects you should determine its position in the sky and assess whether it can be seen or not. This is done in several steps, to facilitate the calculation. First we need to calculate the *hour angle*, t , of each object, which gives its

position in the east-west direction (*remember that the hour angle is measured clockwise from the south*).

1) Compute the sidereal time, Θ (the hour angle, t , of the vernal equinox, Υ)

At noon on March 20 the Vernal Equinox is in the south, so its hour angle, i.e., the sidereal time, is $\Theta = 0^h 0^m 0^s$ (this is not entirely correct, but it is close enough for our purposes).

- i) Given that the Vernal Equinox shifts toward the west with 4^m each day, compute the sidereal time at noon on 14 February.
- ii) We wish to observe in the early evening. Compute the sidereal time at 18:00 on 14 February.

2) Compute the hour angle, t , of the star

The hour angle of a star gives the time that has passed since the star crossed the meridian. At high latitudes this can be used to estimate its position in the sky in the following way: 0^h =South, 6^h =West, 12^h =North and 18^h =East. The hour angle of a star can be found from the difference between the hour angle of the Vernal Equinox and the right ascension, α , of the star: $t = \Theta - \alpha$. Using the values in the table below, and your computed value of Θ , determine the hour angle, t , and corresponding directions for each of the objects in the table below.

Now we know in which direction of the sky to look for the star, but we still do not know how high in the sky it is (its altitude, h).

3) Compute the altitude, h , of the star

Our next step is therefore to estimate this. Doing an exact calculation is complicated, so we will instead make an estimate of the approximate altitude based on the highest and lowest altitude, and the hour angle we computed above. We will do this in several steps. The table below gives the declination, δ , of each object, i.e., its height above the celestial equator.

- i) Show, by drawing or some other method, that the maximum height of a star above the horizon is: $h^* = \delta + 90^\circ - \text{lat}$
- ii) Use this to find a relation between the declination of a star and its minimum height above (or below) the horizon.
- iii) Use these relations to compute the highest and lowest position in the sky for each object in the table below.
- iv) Since an object is highest in the sky when it is in the south, and lowest in the north, corresponding to hour angles of 0 and 12, you should now be able to make an estimate of the position of each object based on the hour angle you computed earlier.

Your task is now to determine the position in the sky for each of the objects in the table (given for February 14 at 18:00) and to discuss whether or not you would expect to be able to see the object during the night of the observation. If the object is not visible at 18:00, do you expect it to be visible at some other time of the night? Why/why not?

Possible observing targets

Below are given the coordinates of a few bright and/or interesting objects. Your task is to convert this to a position in the sky. For the solar system objects the coordinates are given for the time February 14 at 18:00 local time.

Name	Type of object	Right ascension (α)	Declination (δ)
Sirius	Star	6 ^h 46 ^m 16 ^s	- 16° 45'
Mimosa	Star	12 ^h 49 ^m 14 ^s	- 59° 50'
Albireo	Double star	19 ^h 31 ^m 43 ^s	+28° 00'
Ring nebula	Nebula	18 ^h 54 ^m 30 ^s	+33° 03'
Andromeda	Galaxy	0 ^h 44 ^m 05 ^s	+41° 24'
The moon	Moon	19 ^h 40 ^m 04 ^s	-25° 47'
Venus	Planet	22 ^h 30 ^m 52 ^s	-10° 54'
Jupiter	Planet	07 ^h 09 ^m 48 ^s	+22° 47'
Mars	Planet	21 ^h 21 ^m 06 ^s	+16° 37'

Part 2.3: Check your calculations with Stellarium

To check your calculations you can use a planetarium program like Stellarium.

1. Download Stellarium from <http://www.stellarium.org>.
2. Open Stellarium.
3. When you start Stellarium it will be set to the present date and time. Change the location so that it is set to Stockholm.
4. Which stars and planets, if any, are visible?
5. Which time does the Sun set?
6. At what time will Sirius be highest in the sky?
7. Change the time to one hour after sunset. Which stars/planets are visible?
8. Change the day, one day at a time, until one month has passed. What do you observe?
9. If you wish to observe Venus on 1 April, what time is best for this observation?

Part 3: Report

Write a report on your observations and calculations.

Your report should include the following:

- Your name
- Answers to the review questions in part 2.1
- The logbook from your observations
- Calculations of the positions of the objects in the table

- A discussion about which objects will be visible, and why
- Answers to the questions about Stellarium
- A short reflection, $\frac{1}{2}$ -1 page, on your own observations in light of the calculations.

The report should be submitted individually on Canvas no later than 15 May 2026.